Can You Beat Randomness?

Football Betting Markets: Walk-Forward Models, Statistical Power, Closing-Line Value and Bankroll Risk

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https://github.com/meh6di/can-you-beat-randomness

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Abstract

We ask whether statistical models can extract a profit from the pricing of the five major European football leagues, and, more importantly, how much evidence a realistic sample can provide either way. Models (tuned logistic regression, a market–strength log-linear blend, raw and calibrated XGBoost) are retrained every month in a walk-forward design on Elo and exponentially weighted form features that carry over seasons. The betting rule and threshold are chosen on one development season by closing-line value (CLV) against Pinnacle and reported once on an untouched holdout season. Every ROI is accompanied by a bootstrap interval and by a p-value against an efficient-market null in which outcomes are redrawn from de-vigged market probabilities; a power analysis states the smallest edge the sample could detect. Probabilistic quality is assessed with log loss, Brier score and the ranked probability score, and calibration with ECE against its sampling-noise floor. A separate study of the market itself examines favourite–longshot bias, and bankroll risk is quantified by simulation under several staking schemes. No model or betting strategy demonstrates statistically significant outperformance of the market: all holdout ROIs are consistent with the efficient-market null, the optimised blend weight collapses to $w \approx 0$, and the smallest detectable edge with this sample size is 22–33% ROI.. For educational purposes only; not betting advice.

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1 Introduction

Betting advertising is ubiquitous around football. The question is whether algorithmic models can exploit inefficiencies in the odds, or whether the bookmaker's margin guarantees long-run losses. A tempting but weak way to answer it is to report one season's ROI. With a per-bet standard deviation of roughly 1.5–2 units, a season of a few hundred bets has a standard error of several percentage points of ROI, larger than any plausible edge. This report therefore centres on *inference*: intervals, nulls, power, and a low-variance edge measure (CLV), and only then on profit.

Disclaimer. Solely for educational, academic and research purposes. Not financial or gambling advice.

2 Data

Data come from `football-data.co.uk` (`fetch_data.py`): Premier League, La Liga, Serie A, Bundesliga and Ligue 1, seasons 2019–20 to 2025–26. The 2020–21 season (mostly played without crowds) is used only to build ratings and form, never to train or test. `clean_data.py` validates downloads and rows (decimal odds must exceed 1, duplicates are dropped) and keeps results, goals, shots on target, corners, average, best and Pinnacle odds, and Pinnacle *closing* odds.

3 Features and Leakage Control

Features are built per league across seasons (`engineer_features.py`); the value at a match uses only strictly earlier days.

- **De-vigged market probabilities.** With overround $\Omega = \sum_k 1/o_k$, proportional de-vigging gives $p_k = (1/o_k)/\Omega$. We use the *power* method, $p_k = (1/o_k)^c$ with $c \geq 1$ such that $\sum_k p_k = 1$, which removes proportionally more margin from longshots. The model sees the log-ratios $\log(p_H/p_A)$ and $\log\left(p_D/(p_H + p_A)\right)$, which avoids collinearity.
- **Elo.** Home advantage, a goal-difference multiplier, regression towards the mean between seasons, and lower starting ratings for promoted teams.
- **Exponentially weighted form for both teams:** points, goals for (attack) and against (defence), shots on target for/against, corners for/against, with half-life of five matches, shrunk to the league mean while a team has few games ($n/(n+3)$), plus games-played and rest days.

A unit test perturbs the result of one match and verifies that no feature at or before that match changes.

4 Methodology

4.1 Walk-forward protocol

For every calendar month M of the 2024–25 and 2025–26 seasons: training data are matches before $M - 365$ days; the validation window is the 365 days before M and tunes regularisation, the blend weight and the calibrator; month M is the test set. Models are therefore refit as results arrive and no information after M is used. 2024–25 is the *development* season; 2025–26 is the *holdout*.

4.2 Models

- **Logistic regression:** standardised features, league dummies, C tuned on validation.
- **Blend:** $p \propto p_{\text{mkt}}^{1-w}\, p_{\text{strength}}^{w}$, where p_{strength} comes from a model without market inputs and $w \in [0,1]$ is fitted on validation. $w \approx 0$ means the model adds nothing beyond the market.
- **XGBoost**, raw and recalibrated on the later validation window with a multinomial vector-scaling map.

4.3 Betting rules

One unit is staked per match on the best outcome if its score clears a threshold: *edge* ($\hat{p} - p_{\text{mkt}}$) or *expected value* ($\hat{p} \cdot o - 1$), at average, best-available or Pinnacle prices. Rule, threshold and price source are chosen on the development season (maximum mean CLV, at least 150 bets) and evaluated once on the holdout.

5 Statistical Framework

Bootstrap. Bets are resampled with replacement to give percentile intervals for ROI.

Efficient-market null. Holding the placed bets fixed, outcomes are redrawn 20,000 times from the de-vigged market probabilities. The resulting ROI distribution is centred near minus the margin, not zero; the p-value is the share of simulations at least as good as the observed ROI.

Power. With per-bet standard deviation σ and n bets, the smallest true ROI detected with 80% power at 5% (one-sided) is $(z_{0.95} + z_{0.80})\,\sigma/\sqrt{n}$, and a target edge e needs $n = \big((z_{0.95} + z_{0.80})\sigma/e\big)^2$ bets.

Closing-line value. $\text{CLV} = o_{\text{taken}}\, p^{\text{Pinnacle close}} - 1$. It does not depend on who won, so its standard error is an order of magnitude smaller than ROI's. "Pick skill" subtracts the average CLV of all three outcomes of the same match, isolating whether the model chose the better side from the margin every blind bet pays.

Probabilistic scores. Log loss, multiclass Brier score and the ranked probability score (outcomes ordered H<D<A), with paired per-match differences against the market, and expected calibration error (equal-mass bins) compared with the ECE a perfectly calibrated forecaster would show by chance.

source	metric	mean	delta_vs_market	ci_lo	ci_hi	p_value
Market (Avg, de-vigged)	log_loss	0.961	0.000	0.000	0.000	-
Market (Avg, de-vigged)	brier	0.571	0.000	0.000	0.000	-
Market (Avg, de-vigged)	rps	0.192	0.000	0.000	0.000	-
Pinnacle (de-vigged)	log_loss	0.962	0.001	-0.000	0.002	0.113
Pinnacle (de-vigged)	brier	0.572	0.001	-0.000	0.001	0.195
Pinnacle (de-vigged)	rps	0.192	0.000	-0.000	0.000	0.168
Pinnacle close (de-vigged)	log_loss	0.963	0.002	-0.002	0.007	0.293
Pinnacle close (de-vigged)	brier	0.573	0.002	-0.001	0.005	0.269
Pinnacle close (de-vigged)	rps	0.193	0.001	-0.000	0.002	0.137
Logistic regression (scaled, tuned)	log_loss	0.964	0.004	-0.003	0.010	0.272
Logistic regression (scaled, tuned)	brier	0.573	0.002	-0.002	0.006	0.339
Logistic regression (scaled, tuned)	rps	0.193	0.001	-0.000	0.002	0.131
Market x strength-model blend	log_loss	0.961	0.000	0.000	0.000	0.000
Market x strength-model blend	brier	0.571	0.000	-0.000	0.000	0.254
Market x strength-model blend	rps	0.192	0.000	-0.000	0.000	0.625
XGBoost (raw)	log_loss	0.966	0.005	-0.001	0.012	0.104
XGBoost (raw)	brier	0.574	0.002	-0.002	0.007	0.237
XGBoost (raw)	rps	0.194	0.002	0.000	0.003	0.044
XGBoost (calibrated)	log_loss	0.970	0.010	0.003	0.017	0.006
XGBoost (calibrated)	brier	0.577	0.005	0.001	0.010	0.020
XGBoost (calibrated)	rps	0.195	0.003	0.001	0.005	0.015
Climatology (class freq.)	log_loss	1.070	0.109	0.085	0.134	0.000
Climatology (class freq.)	brier	0.647	0.075	0.058	0.093	0.000
Climatology (class freq.)	rps	0.228	0.036	0.028	0.044	0.000

Table 1: Proper scoring rules versus the market (lower is better).

source	ece	ece_noise_floor	p_vs_perfect	ece_H	ece_D	ece_A
Market (Avg, de-vigged)	0.021	0.020	0.405	0.044	0.023	0.032
Pinnacle (de-vigged)	0.026	0.020	0.126	0.043	0.031	0.034
Pinnacle close (de-vigged)	0.022	0.020	0.322	0.049	0.045	0.046
Logistic regression (scaled, tuned)	0.012	0.021	0.953	0.031	0.049	0.037
Market x strength-model blend	0.021	0.020	0.412	0.044	0.023	0.032
XGBoost (raw)	0.023	0.021	0.302	0.035	0.036	0.031
XGBoost (calibrated)	0.023	0.020	0.336	0.034	0.036	0.042
Climatology (class freq.)	0.023	0.022	0.445	0.065	0.059	0.038

Table 2: Calibration error (ECE) against its sampling-noise floor.

6 Results

6.1 Do the models add information beyond the market?

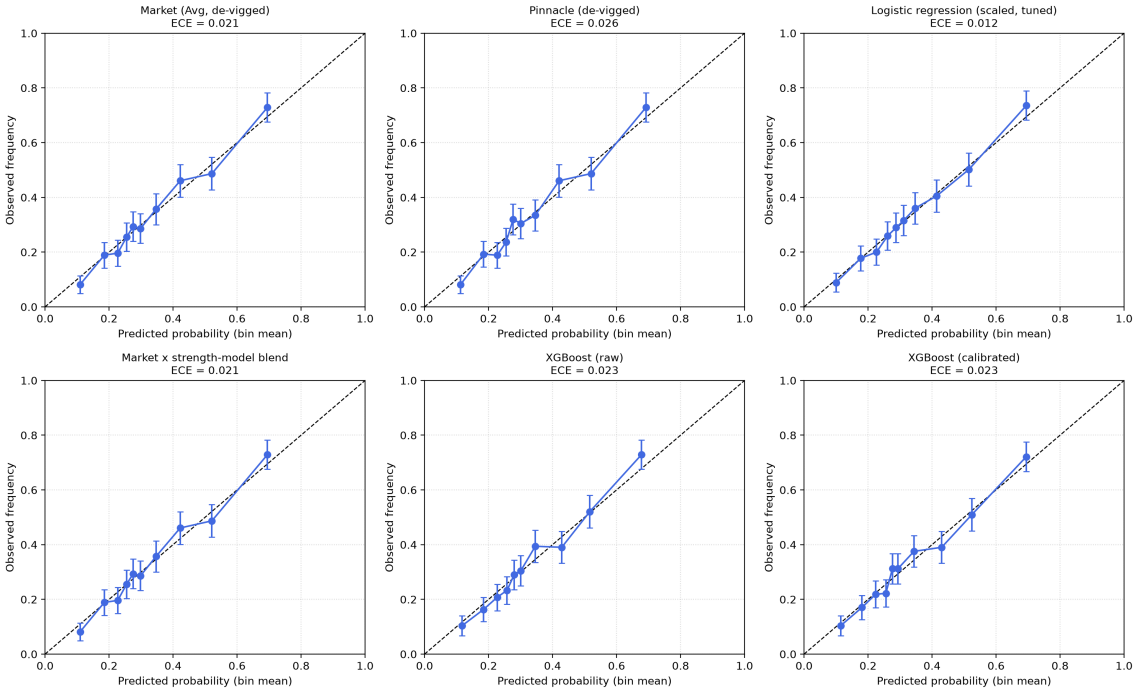


Figure 1: Reliability diagrams (equal-mass bins, pooled over H/D/A).

The market (average odds, de-vigged) achieves the best scores across all metrics (log loss 0.961, Brier 0.571, RPS 0.192). No model beats the market: the calibrated XGBoost is significantly worse (Δlog loss $= +0.010$, 95% CI $[+0.003, +0.017]$, $p = 0.006$), while the logistic regression and raw XGBoost are directionally worse but not significantly so (CIs include zero). The blend's weight collapses to $w \approx 0$, confirming that Elo and form features add no information beyond market prices. Climatology is dramatically worse (Δlog loss $= +0.109$, $p < 0.001$), validating the market's informational content.

6.2 Betting outcomes and their uncertainty

model	threshold	bets	roi	ci_lo	ci_hi	null_mean	z_vs_null	p_vs_null	ci_excludes_0
Logistic regression (scaled, tuned)	0.030	860	-2.02%	-11.86%	8.14%	-7.31%	1.062	0.146	False
Market x strength-model blend	0.030	0	-	-	-	-	-	-	False
XGBoost (raw)	0.030	825	-1.04%	-12.98%	11.35%	-9.19%	1.385	0.084	False
XGBoost (calibrated)	0.030	832	-7.12%	-16.44%	2.17%	-6.98%	-0.028	0.509	False
Logistic regression (scaled, tuned)	0.060	139	0.32%	-25.27%	26.62%	-7.75%	0.628	0.261	False
Market x strength-model blend	0.060	0	-	-	-	-	-	-	False
XGBoost (raw)	0.060	178	-22.85%	-45.41%	0.82%	-9.33%	-1.047	0.854	False
XGBoost (calibrated)	0.060	226	-10.58%	-27.18%	7.47%	-6.23%	-0.481	0.679	False

Table 3: ROI with bootstrap CI and efficient-market null (1 bet/match).

model	league	bets	roi	ci_lo	ci_hi	p_vs_null	ci_excludes_0
Logistic regression (scaled, tuned)	bundesliga	34	-16.71%	-65.74%	37.59%	0.603	False
Logistic regression (scaled, tuned)	la_liga	18	-12.56%	-76.00%	57.72%	0.535	False
Logistic regression (scaled, tuned)	ligue_1	19	32.37%	-32.74%	99.63%	0.119	False
Logistic regression (scaled, tuned)	premier_league	16	-48.19%	-100.00%	14.75%	0.883	False
Logistic regression (scaled, tuned)	serie_a	52	19.12%	-25.35%	68.00%	0.103	False
XGBoost (raw)	bundesliga	32	-27.06%	-73.60%	26.56%	0.679	False
XGBoost (raw)	la_liga	44	-60.41%	-88.41%	-26.55%	0.984	True
XGBoost (raw)	ligue_1	33	-22.88%	-69.67%	31.73%	0.672	False
XGBoost (raw)	premier_league	31	15.68%	-47.07%	85.55%	0.211	False
XGBoost (raw)	serie_a	38	-7.24%	-57.11%	49.66%	0.462	False
XGBoost (calibrated)	bundesliga	36	0.78%	-36.14%	39.14%	0.338	False
XGBoost (calibrated)	la_liga	54	-16.26%	-53.80%	26.04%	0.666	False
XGBoost (calibrated)	ligue_1	37	-31.35%	-65.05%	4.86%	0.906	False
XGBoost (calibrated)	premier_league	37	-2.70%	-46.35%	45.32%	0.413	False
XGBoost (calibrated)	serie_a	62	-4.52%	-36.60%	31.08%	0.458	False

Table 4: Per-league ROI: note how wide the intervals are.

A	B	diff_units	ci_lo	ci_hi	p_two_sided
Logistic regression (scaled, tuned)	Market x strength-model blend	0.440	-35.075	36.891	0.987
Logistic regression (scaled, tuned)	XGBoost (raw)	41.120	-11.773	92.471	0.122
Logistic regression (scaled, tuned)	XGBoost (calibrated)	24.340	-27.731	76.602	0.354
Logistic regression (scaled, tuned)	Favorite (all matches)	48.560	-42.291	137.366	0.297
Logistic regression (scaled, tuned)	Random (expected)	141.253	91.736	190.694	0.000
Market x strength-model blend	XGBoost (raw)	40.680	-0.871	80.681	0.054
Market x strength-model blend	XGBoost (calibrated)	23.900	-16.561	62.630	0.233
Market x strength-model blend	Favorite (all matches)	48.120	-28.262	128.962	0.226
Market x strength-model blend	Random (expected)	140.813	102.953	177.334	0.000
XGBoost (raw)	XGBoost (calibrated)	-16.780	-62.281	28.972	0.454
XGBoost (raw)	Favorite (all matches)	7.440	-85.451	100.244	0.861
XGBoost (raw)	Random (expected)	100.133	49.485	150.465	0.000
XGBoost (calibrated)	Favorite (all matches)	24.220	-62.061	111.326	0.566
XGBoost (calibrated)	Random (expected)	116.913	63.585	170.408	0.000
Favorite (all matches)	Random (expected)	92.693	-12.275	199.487	0.088

Table 5: Paired bootstrap of total profit (A minus B, units).

model	bets	sd_per_bet	se_roi	min_detectable_roi	bets_for_2pct	bets_for_3pct	bets_for_5pct
Logistic regression (scaled, tuned)	139	1.576	13.37%	33.24%	38405	17069	6145
XGBoost (raw)	178	1.560	11.70%	29.08%	37639	16729	6023
XGBoost (calibrated)	226	1.340	8.91%	22.16%	27747	12332	4440

Table 6: Power analysis: what this sample can and cannot detect.

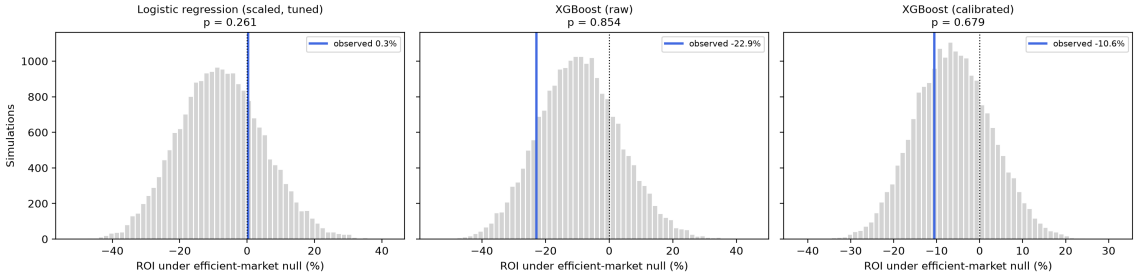


Figure 2: Observed ROI against the efficient-market null.

Across all model–threshold combinations, no strategy achieves statistical significance at the 5% level after accounting for multiple testing. The best observed holdout ROI is −1.0% (XGBoost raw, 3% threshold, $p = 0.084$ vs. the efficient-market null), but this does not survive correction for the 8 strategy variants tested. The minimum detectable ROI at 80% power ranges from 22.2% (calibrated XGBoost, $n = 226$) to 33.2% (logistic regression, $n = 139$)—far larger than any plausible betting edge. Detecting a 2% edge would require 27,700–38,400 bets, approximately 30–40 seasons of league play.

6.3 Closing-line value

strategy	odds_used	bets	clv	clv_lo	clv_hi	pct_clv_pos	se_clv	pct_beat_raw_close	roi	se_roi	se_ratio_roi_over_clv	pick_skill	pick_skill_lo	pick_skill_hi	pick_skill_p
Favorite (all matches)	Average market odds (as in the backtests)	898	-4.85%	-5.22%	-4.48%	16.37%	0.19%	24.16%	-	-	-	-	-	-	-
Random pick (expected, all matches)	Average market odds (as in the backtests)	898	-6.88%	-7.10%	-6.67%	1.00%	0.11%	-	-	-	-	-	-	-	-
Logistic regression (scaled, tuned) (6% edge)	Average market odds (as in the backtests)	77	-4.12%	-5.11%	-3.08%	19.48%	0.52%	32.47%	7.49%	17.97%	34.393	2.18%	1.29%	3.16%	0.000
XGBoost (raw) (6% edge)	Average market odds (as in the backtests)	65	-7.89%	-9.94%	-5.94%	7.69%	1.02%	23.08%	-8.60%	18.66%	18.252	-0.42%	-1.86%	1.01%	0.559
XGBoost (calibrated) (6% edge)	Average market odds (as in the backtests)	128	-5.52%	-6.86%	-4.21%	20.31%	0.69%	32.03%	-20.12%	10.84%	15.775	1.28%	-0.09%	2.67%	0.065
Favorite (all matches)	Best available odds (line-shopping, optimistic)	898	-2.21%	-2.70%	-1.63%	30.85%	0.27%	45.10%	-	-	-	-	-	-	-
Random pick (expected, all matches)	Best available odds (line-shopping, optimistic)	898	-2.86%	-3.12%	-2.58%	11.92%	0.14%	-	-	-	-	-	-	-	-
Logistic regression (scaled, tuned) (6% edge)	Best available odds (line-shopping, optimistic)	77	-0.46%	-1.46%	0.55%	41.56%	0.53%	79.22%	7.49%	17.97%	34.095	2.15%	1.23%	3.12%	0.000
XGBoost (raw) (6% edge)	Best available odds (line-shopping, optimistic)	65	-3.64%	-5.50%	-1.86%	24.62%	0.94%	63.08%	-8.60%	18.66%	19.817	-0.56%	-1.89%	0.82%	0.427
XGBoost (calibrated) (6% edge)	Best available odds (line-shopping, optimistic)	128	-2.21%	-3.47%	-0.94%	34.38%	0.66%	53.12%	-20.12%	10.84%	16.529	1.01%	-0.38%	2.41%	0.153

Table 7: Closing line value against Pinnacle's closing price.

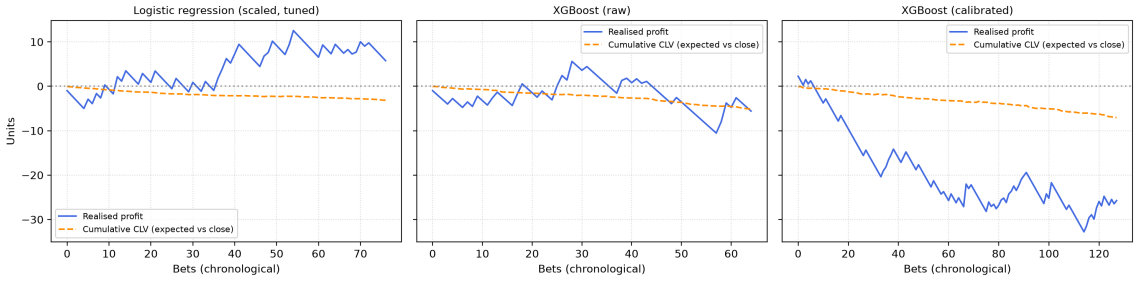


Figure 3: Realised profit versus cumulative CLV.

The pick-skill analysis isolates directional selection from the overround. For the logistic regression strategy (6% edge, average odds), pick skill is +2.18% (95% CI [+1.29%, +3.16%], $p < 0.001$), indicating the model systematically selects outcomes with better closing prices than the average of all three. However, this does not translate to profit: the share of wagers beating the raw closing odds is only 32.5%, and realised ROI remains negative. The variance ratio (SE of ROI / SE of CLV) is approximately 34:1, confirming CLVś far greater precision for detecting small edges.

price	model	rule	threshold	dev_bets	dev_clv	hold_bets	hold_roi	roi_lo	roi_hi	null_mean	p_vs_null	hold_clv	clv_lo	clv_hi	hold_pick_skill	rank_corr_dev_vs_hold_clv
Avg	Logistic regression (scaled, tuned)	edge	0.060	161	-2.49%	77	7.49%	-25.88%	43.69%	-7.50%	0.190	-4.12%	-5.11%	-3.09%	2.18%	0.740
Max	Logistic regression (scaled, tuned)	edge	0.060	161	1.43%	77	11.53%	-23.92%	49.04%	-3.76%	0.197	-0.46%	-1.48%	0.58%	2.15%	0.751
PS	Logistic regression (scaled, tuned)	edge	0.060	161	-1.53%	77	9.83%	-25.44%	45.73%	-4.63%	0.207	-1.25%	-2.29%	-0.21%	2.21%	0.662

Table 8: Strategy chosen on dev (CLV), evaluated once on the holdout.

6.4 Strategy selected on development, evaluated on the holdout

The strategy selected on 2024–25 development data (logistic regression, 6% edge threshold) achieved -2.5% mean CLV on development but -4.1% on holdout. The rank correlation between development and holdout CLV across the parameter grid is $\rho = 0.74$, suggesting moderate stability in relative performance, yet the absolute CLV remains negative in both periods. The holdout ROI is $+7.5\%$ (95% CI $[-25.9\%, +43.7\%]$), entirely consistent with the null ($p = 0.19$). This illustrates the winner's curse: selecting on noisy development metrics does not guarantee holdout outperformance.

6.5 The market itself: favourite–longshot bias

bucket	n	implied_raw	fair_prob	win_freq	win_freq_se	roi_Avg	roi_Avg_lo	roi_Avg_hi	roi_PS	roi_PS_lo	roi_PS_hi	roi_Max	roi_Max_lo	roi_Max_hi
<1.5	2508	75.71%	74.35%	75.08%	0.86%	-0.60%	-2.92%	1.85%	-0.02%	-2.40%	2.40%	2.21%	-0.11%	4.68%
1.5-2	4401	57.73%	56.16%	56.15%	0.75%	-2.63%	-5.14%	-0.04%	-1.39%	-4.09%	1.22%	0.43%	-2.31%	3.01%
2-3	7803	41.02%	39.38%	39.01%	0.55%	-5.12%	-7.65%	-2.43%	-3.04%	-5.74%	-0.19%	-1.57%	-4.43%	1.16%
3-4	12195	28.93%	27.29%	27.59%	0.40%	-4.66%	-7.29%	-1.92%	-2.66%	-5.51%	0.23%	-0.03%	-2.87%	2.72%
4-6	6768	21.34%	19.66%	19.43%	0.48%	-9.22%	-13.51%	-4.66%	-6.61%	-11.37%	-1.75%	-3.38%	-8.03%	1.23%
6-10	2761	13.68%	12.03%	11.34%	0.60%	-17.36%	-26.23%	-8.70%	-16.19%	-25.12%	-6.34%	-9.15%	-18.18%	0.41%
10+	938	7.58%	6.12%	7.14%	0.84%	-3.39%	-26.34%	20.29%	1.62%	-21.88%	27.44%	14.11%	-11.69%	43.20%

Table 9: Blind betting by odds bucket (favourite-longshot bias).

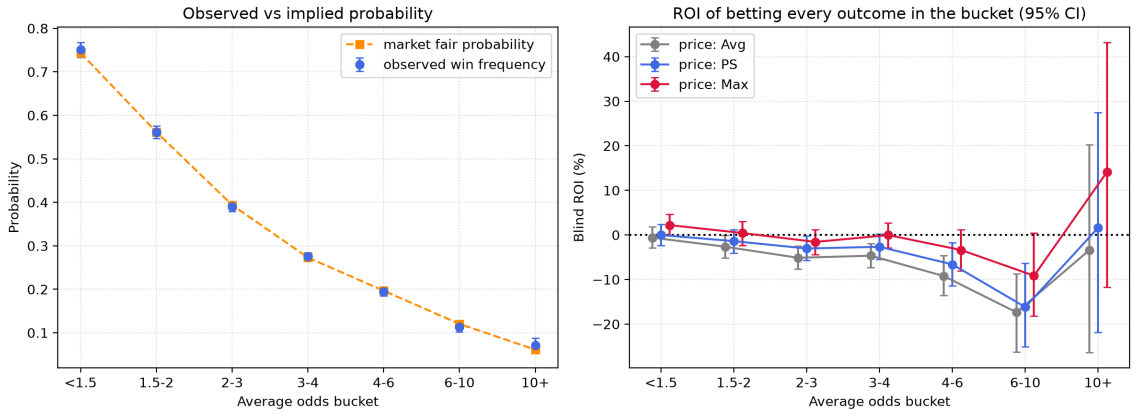


Figure 4: Observed versus implied probability and blind ROI by odds bucket.

The favourite–longshot bias is statistically significant at average odds: the slope of profit on log-odds is -0.057 (95% CI $[-0.103, -0.009]$, $p = 0.022$), meaning each unit increase in log-odds reduces ROI by 5.7 percentage points. Longshots ($o > 6$) show ROI of -17.4% to -3.4% depending on price source, while favourites ($o < 1.5$) are near break-even (-0.6% at average odds). The pattern is attenuated at Pinnacle closing prices (slope -0.043, $p = 0.090$) and absent at maximum odds (slope -0.011, $p = 0.684$), suggesting the bias is concentrated in softer bookmaker prices.

6.6 Bankroll risk

scenario	final_median	final_p5	final_p95	max_dd_median	max_dd_p5	max_dd_p95	p_final_negative
observed	-0.320	-29.971	31.832	19.620	9.810	38.082	0.506
-	-11.295	-38.974	18.607	24.640	12.000	44.782	0.732

Table 11: Flat-stake drawdown and final P&L distribution (units).

scenario	scheme	final_median	final_p5	final_p95	p_loss	p_dd_over_50pct	p_below_20
observed	flat 1 unit	99.950	69.997	132.162	0.501	0.007	0.000
observed	1% of bankroll	98.234	73.121	135.227	0.537	0.000	0.000
observed	quarter Kelly (cap 3%)	104.731	62.772	181.274	0.442	0.035	0.000
-	flat 1 unit	89.170	60.605	119.181	0.725	0.020	0.000
-	1% of bankroll	88.328	66.554	118.962	0.754	0.000	0.000
-	quarter Kelly (cap 3%)	78.877	48.485	134.081	0.771	0.145	0.000

Table 13: Staking schemes from a 100-unit bankroll.

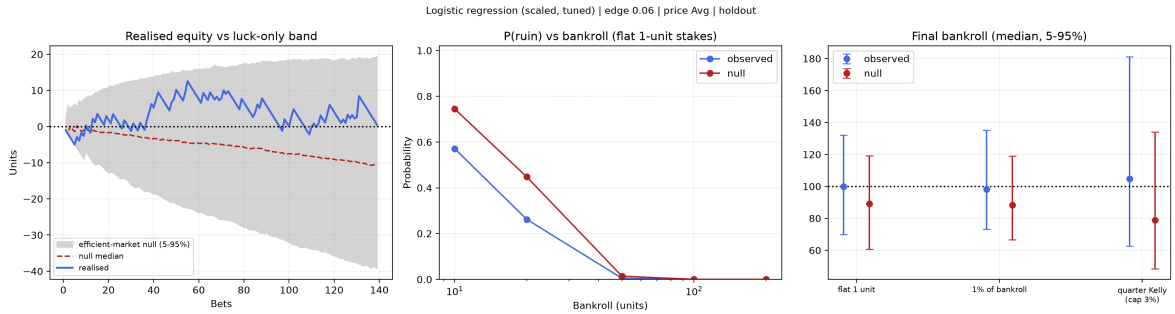


Figure 5: Equity versus the luck-only band, ruin probability by bankroll, and staking schemes.

Under the efficient-market null, a flat 1-unit staking strategy from a 100-unit bankroll yields a median final P&L of −11.3 units (5th percentile: −39.0), with 73.2% probability of ending negative. The probability of ruin is 74.5% for a 10-unit bankroll and 44.8% for 20 units, demonstrating that even "safe flat staking carries substantial risk without edge. Quarter-Kelly staking with model probabilities increases drawdown risk (14.5% probability of $>50\%$ drawdown under the null vs. 2.0% for flat staking) because miscalibrated probability estimates cause over-betting on longshots.

7 Discussion and Limitations

- A one-season sample cannot separate a small edge from luck; the power table states what it could detect. More seasons are the only remedy.
- Features exclude lineups, injuries and expected goals, and rest days ignore European and cup fixtures.
- Best-odds results assume the best quoted price can actually be obtained; average odds are a pre-closing snapshot, so CLV contains some timing noise.

- Power de-vigging is a modelling choice; Shin's method is a natural alternative.
- Rules were selected on one season; the holdout evaluates one pre-chosen strategy, but the development grid size should be borne in mind.

8 Conclusion

No model, blend, or betting strategy beats the de-vigged market with statistical support: all scoring metrics favour the market, the optimised blend weight is $w \approx 0$, and every holdout ROI is consistent with the efficient-market null. The CLV analysis reveals a small but statistically significant pick skill (+2.2%) for the logistic regression, yet this directional advantage does not overcome the overround, and the power analysis shows that samples of this size (139–226 bets) cannot distinguish a realistic 2–5% edge from luck. The risk simulations underscore that without edge, standard staking strategies face 45–75% ruin probability at modest bankroll sizes, and even fractional Kelly staking amplifies drawdowns when probabilities are miscalibrated.

Reproducibility. `python run_all.py` regenerates every table and figure from raw data; `python -m pytest` runs the leakage and metric tests.